

6 All living organisms are made of cells.

(a) The table shows some structures found in living cells.

Complete the table to show in which type of cells these structures are present (✓) or absent (X).

(4)

Structure	Animal cell ✓ = present X = absent	Plant cell ✓ = present X = absent	Prokaryotic cell ✓ = present X = absent
amyloplast			
circular DNA			
mitochondria			
nucleolus			

(b) The smallest living cell has a diameter of $0.2\text{ }\mu\text{m}$.

The volume of a sphere is calculated using the formula:

$$\frac{4}{3}\pi r^3$$

Which of the following shows the volume of this cell in standard form?

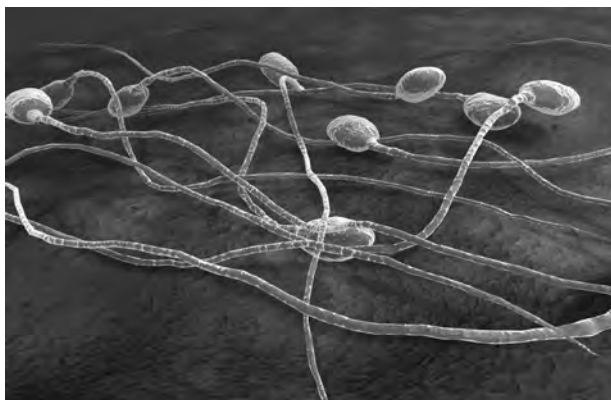
(1)

- ☐ **A** $4.2 \times 10^{-3}\text{ }\mu\text{m}^3$
- ☐ **B** $4.2 \times 10^3\text{ }\mu\text{m}^3$
- ☐ **C** $3.4 \times 10^{-2}\text{ }\mu\text{m}^3$
- ☐ **D** $3.4 \times 10^2\text{ }\mu\text{m}^3$



(c) Gamete cells are specialised for their function.

(i) The photograph shows some sperm cells, as seen using a microscope.



×4000

(Source: © MedicalRF.com/Alamy Stock Photo)

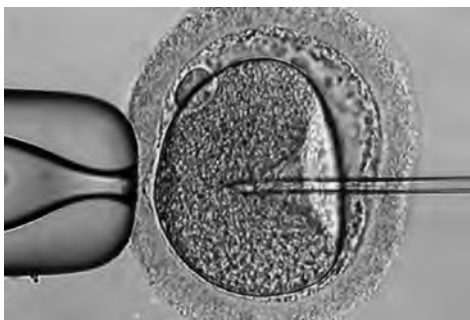
Which of the following statements about human sperm cells are correct?

1. they contain the same quantity of DNA as a body cell
2. they contain hydrolytic enzymes
3. the release of energy from starch enables the rotation of the flagellum

(1)

- ☐ **A** 1, 2 and 3
- ☐ **B** 2 and 3 only
- ☐ **C** 2 only
- ☐ **D** 3 only

- *(ii) The photograph shows a sperm cell being injected into a human egg cell during the process of in-vitro fertilisation (IVF).



(Source: © Phanie / Alamy Stock Photo)

Scientists have suggested that one cause of human infertility is the production of egg cells with a zona pellucida thickness greater than $18\text{ }\mu\text{m}$.

The table shows the mean thickness of the zona pellucida of fertilised egg cells and unfertilised egg cells.

Type of egg cell	Mean zona pellucida thickness / μm
fertilised	16.6 ± 3.2
unfertilised	18.9 ± 4.0

Sperm cells are released from the male by ejaculation.

Another suggested cause of infertility is a longer time between sperm cell ejaculation and the cells reaching an egg cell in the female.

Scientists wish to increase the number of successfully fertilised egg cells by IVF. They have proposed that the technique of injecting recently obtained sperm cells should be used for egg cells with a zona pellucida thicker than $18\text{ }\mu\text{m}$.

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Evaluate this proposal.

Use the information provided in the question and your own knowledge to support your answer.

(6)

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(Total for Question 6 = 12 marks)

